

Ground Truth: Delay Injection Points

All four video sets with verified delay injection timestamps, visual elements (AOIs), and analysis windows.

Set A — Hill Climbing

Delay Point	Timestamp	Visual Element (AOI)	Analysis Window
1	72.0s	Binary tree diagram	67s-77s
2	112.0s	State Space diagram — "Plateau" word appears	110s-120s
3	150.0s	Simulated Annealing — thermometer icon	147s-157s
4	173.0s	Blue cliff-jump transition slide — jumping man icon	169s-179s

Set B — KNN

Delay Point	Timestamp	Visual Element (AOI)	Analysis Window
1	43.0s	K value — most preferred is 5	38s-48s
2	129.0s	Step 1 — Select the number (K) of the neighbors	124s-134s
3	159.0s	Euclidean distance formula — $d(x,y) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$	154s-164s
4	220.0s	Step 4 — Small=4, Medium=1 classification count	215s-225s

Set C — Bayes Theorem

Delay Point	Timestamp	Visual Element (AOI)	Analysis Window
1	44.0s	Bayes Theorem formula — $P(B A) \times P(A) / P(B)$	39s-49s
2	100.0s	Probability values — $P(\text{Homicidal}) = 0.0001$	95s-105s
3	151.0s	100K Population in circle	146s-156s
4	182.0s	Venn diagram — 1.5 homicidals not gamers / 8.5 homicidals are gamers	177s-187s

Set D — Neural Networks

Delay Point	Timestamp	Visual Element (AOI)	Analysis Window
1	27.0s	Neural network diagram — b1 (bias) in hidden node	22s-32s
2	110.0s	ReLU activation function graph (0.571)	105s-115s
3	195.0s	Output node — weighted inputs (0.571)(0.704)(0.812) with Sigmoid	190s-200s

4	212.0s	Sigmoid curve — input 1.681 mapping to output 0.843	207s–217s
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Notes

Baseline period: 2-second window immediately before each delay point onset (e.g., if delay is at 72s, baseline = 70–72s).

Each video set has 4 variants: X0 (no delay), X1 (1s delay), X3 (3s delay), X5 (5s delay).

Delays are implemented as freeze-frames — the video frame is held static while audio continues, keeping total video duration constant.